NEW HORIZON COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning

Miniproject Review 1
(24AIM57)

SPILLSENTINEL - Marine Oil-Spill Attribution, Forecasting and Impact Assessment System

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Agenda

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- Introduction
- Objective
- Scope
- Literature Survey
- Existing System
- Proposed Methodology
- Mathematical Model
- Conclusion
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ABSTRACT

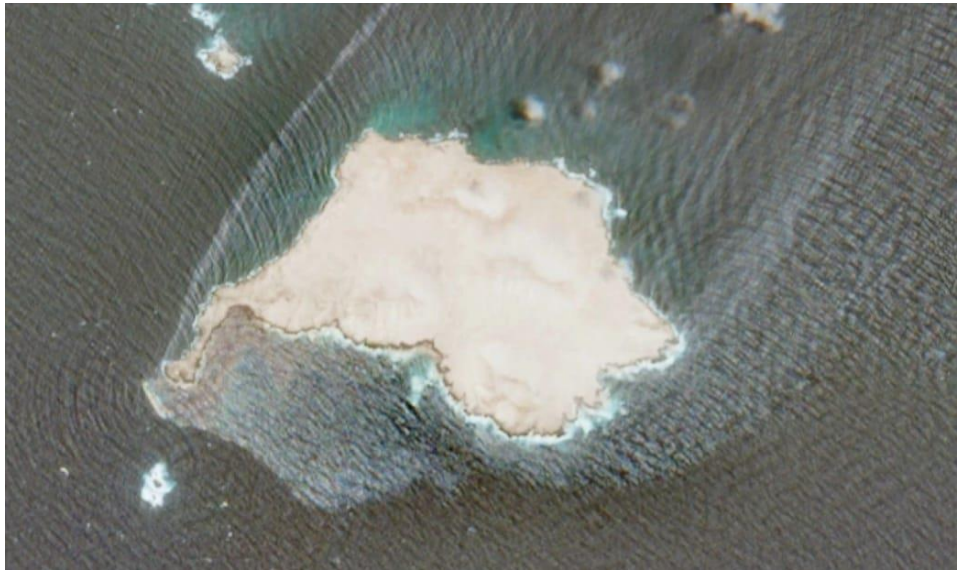
- SpillSentinel is an AI-based system for detecting and investigating marine oil spills using SAR imagery, oceanographic data, and AIS information.
- The system estimates the probable spill origin and responsible vessel while considering SAR look-alikes, AIS reliability, and uncertainty in the analysis.
- It further forecasts the probable spread of the spill and assesses potential impact on Marine Protected Areas.
- The proposed workflow provides an integrated GIS-based platform for oil-spill investigation and visualization.

Introduction

- Marine oil spills pose a serious threat to marine ecosystems, coastal habitats, fisheries and communities that depend on the ocean.
- **In August 2026, an oil spill from the grounded tanker Caroline Bezengi off Oman expanded from tens of squares kilometers to several hundred square kilometers to several hundred square kilometers within days, reaching coastal area and threatening a marine protected area**
- **The satellite observations showing the affected area expanding from approximately 20 km² to 150 km² within one week.**
- **The spill also reached coastal areas and a protected marine zone, putting sensitive marine biodiversity at risk, including endangered species.**
- However, identifying oil spills from satellite imagery is challenging, as dark SAR regions can result from natural ocean phenomena, while incomplete AIS data and changing ocean conditions complicate vessel attribution and tracking.
- Therefore, there is a need for an integrated system that can validate oil spills, trace their probable origin, correlate them with vessels, forecast their movement and assess potential marine impact.

WORLD NEWS

Oil spill from grounded tanker off Oman expands sharply, satellite images show



1 of 2 | This satellite image released by European Space Agency shows oil leaking from a grounded U.K.-sanctioned tanker near Qibliyah Island off the coast of Oman, Tuesday, July 28, 2026. (European Space Agency via AP) License this photo

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Objective

- To detect and validate marine oil spills using Sentinel-1 SAR imagery.
- To estimate the probable spill origin and release time using drift and oceanographic data.
- To correlate AIS data and assess candidate vessels based on spatial and temporal evidence.
- To improve attribution reliability through AIS reliability analysis, explainable evidence, and uncertainty assessment.
- To forecast the probable future spread of the spill using environmental conditions.
- To assess potential marine impact by identifying overlap with Marine Protected Areas.

Scope Of the Project

- The system focuses on detecting and validating marine oil spills using Sentinel-1 SAR imagery.
- It estimates the probable spill origin and release time using drift and oceanographic data.
- It correlates AIS data with spill information to assess and attribute candidate vessels.
- It incorporates AIS reliability, explainable attribution, and uncertainty analysis into the investigation.
- It forecasts the probable future spread of the detected oil spill using environmental conditions.
- It assesses potential impact by identifying overlap between the forecasted spill region and Marine Protected Areas.

LITERATURE REVIEW

S.No	Year	Title Of Paper	Author , Conference/Journal	Methodology Used	Gaps Identified/Remarks
1	**2026**	**Detection and discrimination of marine oil spills and look-alike phenomena in synthetic aperture radar imagery**	X. Huang et al., *GIScience & Remote Sensing*	MDODNet for multi-class detection of oil spills and SAR look-alikes using Sentinel-1 imagery.	Strongly addresses **look-alike discrimination**; does not cover vessel attribution or spill-source analysis. (Taylor & Francis Online)
2	**2026**	**Two-Stage Oil Spill Detection in SAR Using a Domain-Adapted Segment Anything Model**	G. Giannopoulos et al., *Remote Sensing*	ConvNeXt-T screening followed by domain-adapted SAM segmentation using Sentinel-1 VV and texture features.	Improves SAR segmentation and look-alike separation; focused mainly on **detection**, not attribution. (MDPI)
3	**2026**	**Automated oil spill detection using deep learning and SAR satellite data: a case study of the Lower Congo Basin**	P. Cheng, Y. Qi, J. Zhao, *Frontiers in Earth Science*	NOS-Net combines U-Net, ResNet and attention mechanisms for Sentinel-1 oil-spill segmentation.	Improves detection of oil spills and look-alikes; remains primarily a **detection-level solution**. (Frontiers)
4	**2026**	**CDANet: A context-detail aware network for marine oil spill detection in SAR imagery**	H. Zhang et al., *Journal of Hazardous Materials*	Context-Detail Aware Network using multi-scale features, attention and a multi-region SAR dataset.	Focuses on high-precision SAR detection; does not extend into **source attribution or forecasting**. (ScienceDirect)

S.No	Year	Title Of Paper	Author , Conference/Journal	Methodology Used	Gaps Identified/Remarks
5	**2025**	**GSCAT-UNET: Enhanced U-Net model with spatial-channel attention gate and three-level attention for oil spill detection using SAR data**	K. Prajapati, M. Bhavsar, A. Mahajan, *Marine Pollution Bulletin*	Enhanced U-Net with spatial-channel attention and multi-level attention for oil-spill and look-alike segmentation.	Directly supports the **look-alike discrimination** research gap. (ScienceDirect)
6	**2025**	**TransOilSeg: A Novel SAR Oil Spill Detection Method Addressing Data Limitations and Look-Alike Confusions**	Y. Chai et al., *IEEE Transactions on Geoscience and Remote Sensing*	Transformer/transfer-learning based SAR segmentation addressing limited data and look-alike confusion.	Addresses data limitations and look-alikes but focuses on **SAR detection**, not vessel attribution. (ResearchGate)
7	**2025**	**Dataset of oil slicks, look-alikes and remarkable SAR signatures obtained from Sentinel-1 data in the Eastern Mediterranean Sea**	Y.-J. Yang, S. Singha, R. Goldman, F. Schütte, *Earth System Science Data*	Provides a labelled Sentinel-1 dataset containing oil slicks, look-alikes and other dark SAR signatures.	Demonstrates the importance of **look-alike datasets and reliable validation** for oil-spill detection. (ESSD)
8	**2025**	**Identification of oil spills in the eastern Arabian Sea using remote sensing and assessment of vulnerable coastal areas using numerical modeling**	V. Trinadha Rao, V. Suneel, *Marine Pollution Bulletin*	Uses remote sensing to identify spills and GNOME to model their fate and movement toward vulnerable coastal areas.	Highly relevant to **spill forecasting and marine-impact assessment**, but does not integrate AIS attribution. (ScienceDirect)
9	**2025**	**A comprehensive analysis of impacts of accidental oil spills emanating from marine traffic in the Eastern Arabian Sea region**	B. Yedla, H. V. R. Mittal, *Regional Studies in Marine Science*	Uses MOHID with met-ocean data to model oil-spill trajectories and identify vulnerable coastal regions.	Strong for **impact assessment**, but separate from SAR detection and vessel attribution. (ScienceDirect)
10	**2024**	**A new ship tracing technology from oil spills based on multi-source data**	D. Luo et al., *Marine Pollution Bulletin*	Combines **SAR, AIS, wind/current data and bidirectional Lagrangian drift modelling** to identify the likely spill vessel.	Extremely relevant to your core attribution system; leaves room for **look-alike validation, uncertainty handling and marine-impact integration**. (ScienceDirect)

Existing System

CleanSeaNet - EMSA

Satellite-based oil spill monitoring
SAR + AIS + Earth Observation

Detects possible pollution and supports vessel identification.

WebGNOME - NOAA

Oil spill trajectory modelling
Wind + Ocean Currents + Drift

Models possible movement and spread of oil.

OOSA - INCOIS

Oil spill trajectory advisory
Oceanographic Data + Modelling

Provides trajectory forecasts for Indian waters.

Maritime Surveillance Systems

Vessel & maritime activity monitoring
AIS + Radar + Satellite Data

Tracks vessel movements and supports maritime surveillance.

Existing solutions support oil-spill monitoring, trajectory modelling, vessel tracking, and maritime surveillance using satellite observations, oceanographic data, and AIS information.

Identified Gap

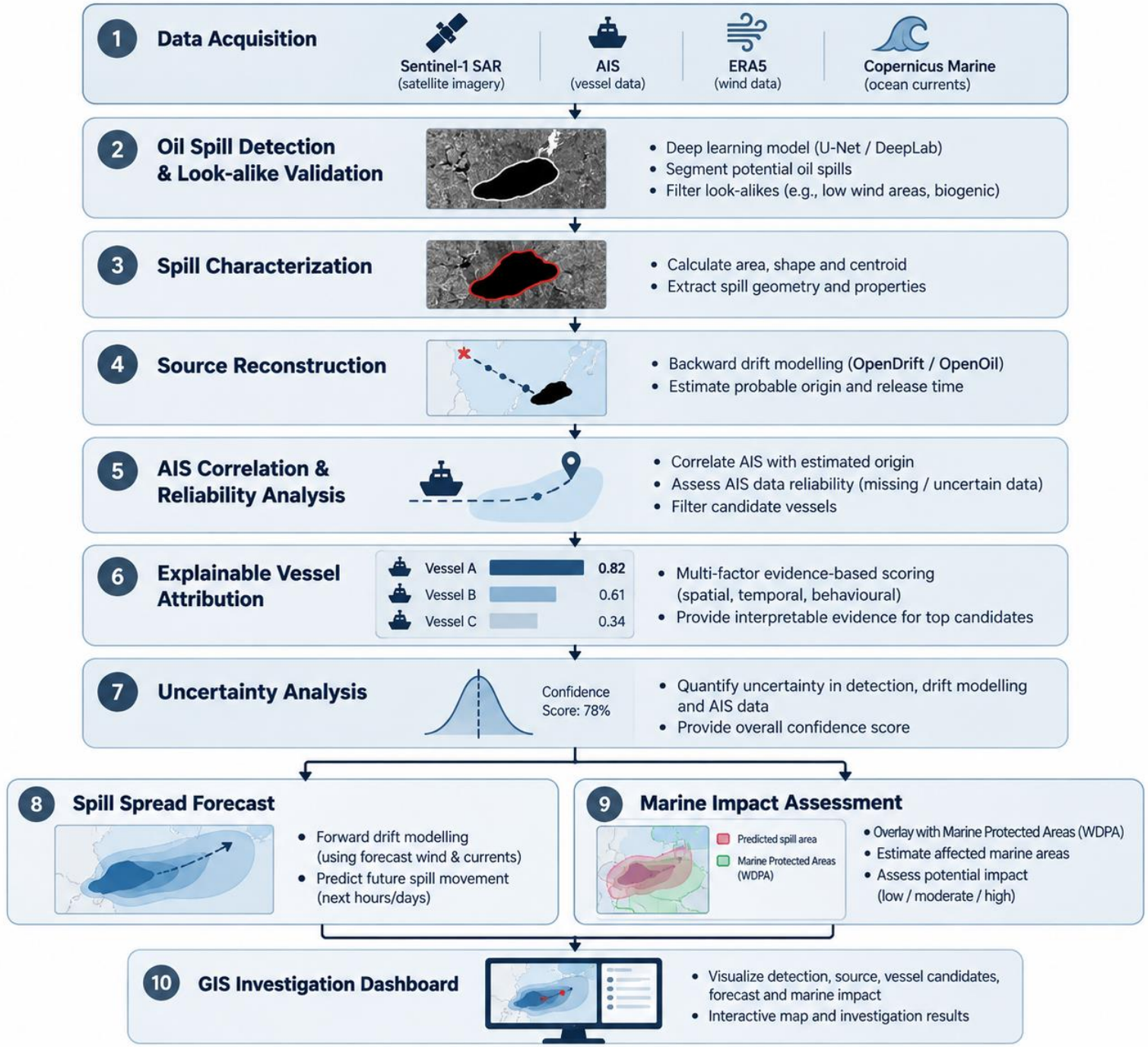
These capabilities are largely distributed across different systems. A unified workflow that connects spill validation, source attribution, AIS reliability, uncertainty analysis, spill forecasting, and marine impact assessment can provide a more complete investigation process.

Proposed Methodology

1. **Data Acquisition**
   Sentinel-1 SAR imagery, AIS trajectories, and wind and ocean-current data are collected for the study region.
2. **Spill Detection & Validation**
   A deep-learning model identifies oil-like regions in SAR imagery, followed by look-alike discrimination to validate the detected spill.
3. **Source Reconstruction**
   The spill is characterized and backward drift modelling is used to estimate its probable origin and release-time window.
4. **Vessel Attribution**
   AIS data is correlated with the estimated origin and time, while AIS reliability and multiple evidence factors are considered for explainable vessel attribution.
5. **Uncertainty & Spill Forecasting**
   Uncertainty from detection, drift, AIS, and attribution is evaluated, followed by forecasting of the spill's probable future movement using forecast environmental conditions.
6. **Marine Impact Assessment**
   The forecasted spill region is compared with Marine Protected Areas to identify potentially affected sensitive marine zones.

Final Output:

- Probable Spill Origin
- Vessel Attribution
- Spill Forecast
- Potential Marine Impact


1 Data Acquisition
Sentinel-1 SAR (satellite imagery)
AIS (vessel data)
ERA5 (wind data)
Copernicus Marine (ocean currents)
2 Oil Spill Detection & Look-alike Validation
• Deep learning model (U-Net / DeepLab)
• Segment potential oil spills
• Filter look-alikes (e.g., low wind areas, biogenic)
3 Spill Characterization
• Calculate area, shape and centroid
• Extract spill geometry and properties
4 Source Reconstruction
• Backward drift modelling (OpenDrift / OpenOil)
• Estimate probable origin and release time
5 AIS Correlation & Reliability Analysis
• Correlate AIS with estimated origin
• Assess AIS data reliability (missing / uncertain data)
• Filter candidate vessels
6 Explainable Vessel Attribution
Vessel A 0.82
Vessel B 0.61
Vessel C 0.34
• Multi-factor evidence-based scoring (spatial, temporal, behavioural)
• Provide interpretable evidence for top candidates
7 Uncertainty Analysis
Confidence Score: 78%
• Quantify uncertainty in detection, drift modelling and AIS data
• Provide overall confidence score
8 Spill Spread Forecast
• Forward drift modelling (using forecast wind & currents)
• Predict future spill movement (next hours/days)
9 Marine Impact Assessment
Predicted spill area
Marine Protected Areas (WDPA)
• Overlay with Marine Protected Areas (WDPA)
• Estimate affected marine areas
• Assess potential impact (low / moderate / high)
10 GIS Investigation Dashboard
• Visualize detection, source, vessel candidates, forecast and marine impact
• Interactive map and investigation results

Conclusion

- SpillSentinel integrates SAR, AIS, and oceanographic data for comprehensive oil-spill investigation.
- The system improves vessel attribution through look-alike validation, AIS reliability analysis, and explainable evidence.
- Spill forecasting extends the system to estimate the future movement of the detected spill.
- Marine impact assessment identifies potentially affected protected marine areas.

Final Outcome

SpillSentinel brings oil-spill detection, vessel attribution, spill forecasting, and marine impact assessment together in one unified system.

References

[1] X. Huang, B. Zhang, et al., “Detection and discrimination of marine oil spills and look-alike phenomena in synthetic aperture radar imagery,” *GIScience & Remote Sensing*, vol. 63, no. 1, 2026.

[2] G. Giannopoulos, M. Kremezi, V. Karathanassi, V. Andronis, D. Bliziotis, et al., “Two-Stage Oil Spill Detection in SAR Using a Domain-Adapted Segment Anything Model,” *Remote Sensing*, vol. 18, no. 12, Art. no. 1948, 2026, doi: 10.3390/rs18121948.

[3] P. Cheng, Y. Qi, and J. Zhao, “Automated oil spill detection using deep learning and SAR satellite data: A case study of the Lower Congo Basin,” *Frontiers in Earth Science*, vol. 13, Art. no. 1667450, 2026, doi: 10.3389/feart.2025.1667450.

[4] Z. Wang, H. Zhang, Z. Guo, Y. Fan, L. Xu, X. Li, and C. Wang, “CDANet: A context-detail aware network for marine oil spill detection in SAR imagery,” *Journal of Hazardous Materials*, vol. 501, Art. no. 140683, 2026, doi: 10.1016/j.jhazmat.2025.140683.

[5] K. Prajapati, M. Bhavsar, and A. Mahajan, “GSCAT-UNET: Enhanced U-Net model with spatial-channel attention gate and three-level attention for oil spill detection using SAR data,” *Marine Pollution Bulletin*, vol. 212, Art. no. 117583, 2025, doi: 10.1016/j.marpolbul.2025.117583.

[6] Y. Chai, X. Han, Y. Wang, D. Luo, J. Yang, P. Chen, and G. Zheng, “TransOilSeg: A novel SAR oil spill detection method addressing data limitations and look-alike confusions,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 63, pp. 1–16, 2025, doi: 10.1109/TGRS.2025.3542810.

[7] Y.-J. Yang, S. Singha, R. Goldman, and F. Schütte, “Dataset of oil slicks, look-alikes and remarkable SAR signatures obtained from Sentinel-1 data in the Eastern Mediterranean Sea,” *Earth System Science Data*, vol. 17, pp. 6807–6837, 2025, doi: 10.5194/essd-17-6807-2025.

[8] V. Trinadha Rao and V. Suneel, “Identification of oil spills in the eastern Arabian Sea using remote sensing and assessment of vulnerable coastal areas using numerical modeling,” *Marine Pollution Bulletin*, vol. 221, Art. no. 118582, 2025, doi: 10.1016/j.marpolbul.2025.118582.

[9] B. Yedla and H. V. R. Mittal, “A comprehensive analysis of impacts of accidental oil spills emanating from marine traffic in the Eastern Arabian Sea region,” *Regional Studies in Marine Science*, vol. 91, Art. no. 104570, 2025, doi: 10.1016/j.rsma.2025.104570.

[10] D. Luo, P. Chen, J. Yang, X. Chen, X. Li, G. Zheng, L. Ren, and Y. Zhao, “A new ship tracing technology from oil spills based on multi-source data,” *Marine Pollution Bulletin*, vol. 207, Art. no. 116808, 2024, doi: 10.1016/j.marpolbul.2024.116808.

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